



PPE Guide



Equipments

**BOUYGUES
CONSTRUCTION**

DISTRIMO

12

HEALTH & SAFETY BASICS



Our 12 Health and Safety Basics are key elements to prevent occupational accidents and protect lives. They are based on feedback and lessons learned. Encompassing the situations most frequently encountered in our construction sites or operations. They are intended to be shared across BYCN's business and operations. They must be fully understood and obeyed by everyone. These rules are intended for the people who organize work, the people who carry it out and the people who control it. They are one component of an appropriate identification and sanction policy.

Personal Protective Equipment

All personnel, including visitors, who work on or visit our construction sites shall wear as a minimum:

- Safety shoes
- A hard hat
- High Visibility Jacket
- Safety glasses
- Hearing protection
- Safety gloves

And task-specific PPE when required as defined by task specific risk assessment

For activities in third party premises, specific PPE must be defined after task risk analysis



Protecting the head

- ▶ Presentation.....6
- ▶ General-purpose protective helmets.....7
- ▶ Special helmets.....8
- ▶ Hats.....10
- ▶ Chin trap.....9



Protecting the hands

- ▶ Presentation.....11
- ▶ Gloves.....12



Protecting the feet

- ▶ Presentation.....20
- ▶ General-purpose high-cut shoes..... 21
- ▶ Special shoes.....22



Protecting the eyes

- ▶ Standards/Recommendations.....26
- ▶ Multi-purpose goggles.....27
- ▶ Specific.....28



Protecting the hearing

- ▶ Moulded ear plugs.....32
- ▶ Disposable ear plugs.....33
- ▶ Earmuffs.....32



Respiratory protection

- ▶ Standards/Recommendations.....34
- ▶ Protective devices.....35

Special work

▶ Harness..... 43

▶ Other.....45



Protections against the weather

▶ Products available.....47



PPE sheets

- ▶ I am a visitor from outside.....48
- ▶ I am an employees.....49
- ▶ I am digging / sanding / sawing.....50
- ▶ I am concreting.....51
- ▶ Asbestos/Lead.....52
- ▶ Chemicals.....53
- ▶ I am an electrician.....54



Pablo codes p.55



PPE Guide - 05/19

WE LOVE LIFE,
WE PROTECT IT.



DISTRIMO



This Guide lists all the PPEs referenced in PABLO that must be supplied and worn at all our work sites. You must go via Distrimo

For special work not shown in this Guide, please contact your Health, Safety and Risk-Prevention Department and DISTRIMO

Personal protective equipment is necessary:

When it is impossible to eliminate the risk

When collective protection is not possible or the risk is too high

The wearer of the PPE must:

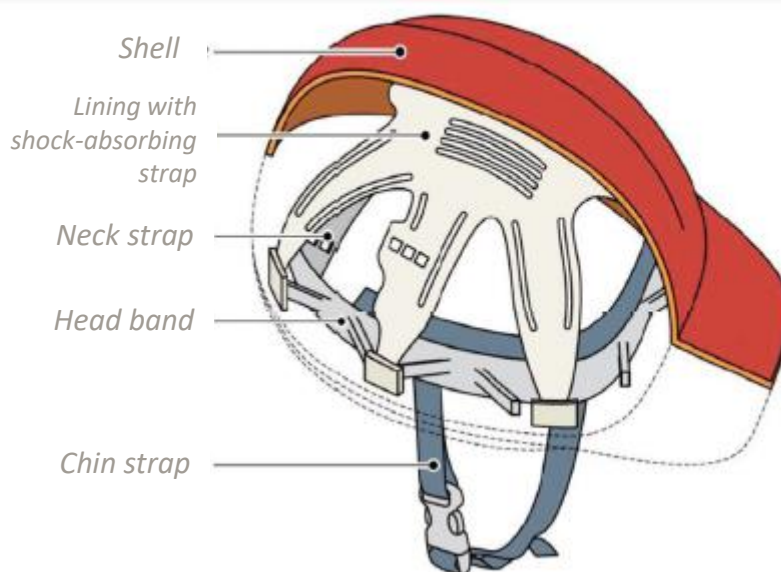
Wear the PPE items supplied

Take care

Ask for replacements if they deteriorate



How does the helmet protect?



The **shell** is designed to withstand external impacts. It is made of a light and strong material such as polyethylene or polycarbonate.

The **cradle** keeps the helmet in place on the head, and acts as a shock absorber during an impact.

The **chin strap** is designed to ensure that the helmet stays on the head, particularly when working at a height.



IRIS II HELMET



IRIS 2 (360g)
(option : safety glasses or badge holder)



FOR WHAT TYPE OF ACTIVITY?

**ALL ACTIVITIES (STRUCTURAL WORK,
SUB-TRADES AND RESTORATION WORK,
CIRCULATING WITHIN A CRANE'S SWING
RADIUS, LOADING / UNLOADING,
STEERING A CRADLE, T&D ETC.)**

KARA HELMET



KARA (340g)
(option : safety glasses
or badge holder)

BRENNUS HELMET



BRENNUS (330g)
(option safety glasses or
label/badge holder)



FOR WHAT TYPE OF ACTIVITY?

VISITORS FROM OUTSIDE



CHIN STRAP MANDATORY FOR ALL





IDRA 2



with shield, no ventilation,
420V, with chin strap with 4
attachment points + chin guard
(650g)

FOR WHAT TYPE OF ACTIVITY?

WORK THAT IS ELECTRICAL IN ORIGIN

HV FITTER

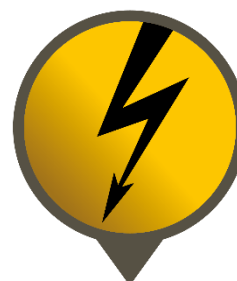
WORK AT SITES NOT AT VOLTAGE

ELECTRICAL WORK /LIVE WORKING

E-MAN



with shield, no ventilation,
1000V
(525g)



with shield, no ventilation, with chin strap with 4
attachment points
1000V
(585g)



KARA



with short visor, chin strap with 4 attachment points, label holder (400g)



FOR WHAT TYPE OF ACTIVITY?

FITTER

WORKING AT HEIGHT

ALVEO BEST



with short visor, chin strap with 4 attachment points, no label/badge holder, no drainage channels (rain) (350g) *no visor = short visor*

CHIN STRAP WITH 4 ATTACHMENT POINTS MANDATORY



chin strap with 4 attachment points in synthetic fabric, attached with quick-release buckle. Suitable for Iris II and Kara helmets.

Style: Kara

Colour: Black

Size: Free size

To avoid any risk of strangulation, the chin strap or its attachment points must yield to a force of around 150-250 N. Four-point chin straps are preferred. The chin strap is not supplied with the helmet (it must be ordered in addition).

A chin strap must be worn in situations where the employee may lose his/her helmet without being able easily to retrieve it: for instance, during work to erect/dismantle metal frameworks and structures (scaffolding) or for work on pylons, etc.





FIRST Base+



grey or navy
50mm visor
(150g)
(no ventilation, OK for electrical work)

FIRST Base 3 XTRA



54mm visor, available in navy and
fluorescent orange
(180g)

FOR WHAT TYPE OF ACTIVITY?

WORKSHOPS

CABLING ON NON-LIVE ELECTRICAL EQUIPMENT

JOBS ON OCCUPIED SITES, FINISHING TRADES. AFTER-SALES SERVICE, INSIDE WORK (GET/FM),

> VALIDATION BY ENTITY

or general-purpose hat (p.8)



PRECAUTIONS: A shock-proof hat protects against injury from a minor impact (e.g. a knock against an immobile object such as the edge of a wall). It is useful in the finishing trades, occupied sites, inside work, or in cramped areas, for instance work in false ceilings. It is not designed to protect against ejected material, falling objects or suspended or moving loads. It is designed for use inside.

In no circumstances is the hat a substitute for a protective helmet. It may not replace the helmet for outside work.

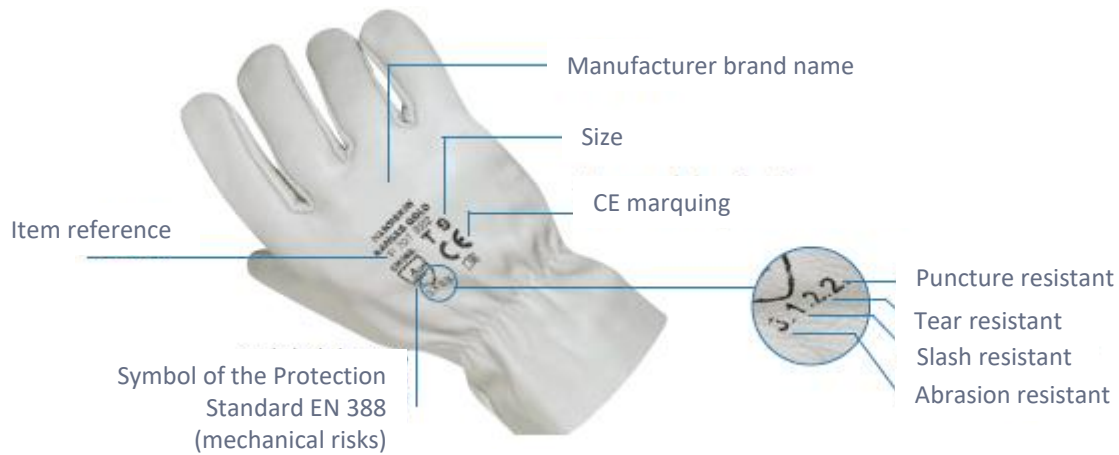


Leather hood

FOR WHAT TYPE OF ACTIVITY?

WELDING WORK





The standard on **anti-cut protective gloves** has changed (ISO 13997 added):

- Better guaranteed anti-cut protection with new tests carried out during manufacture (blade pressure tested in addition to cyclical passes over the glove)
- A new marking on gloves

Abrasion		Tear		Performance letter	NEW
E.g.	4	3	4	3	C (P)
Cutting		Puncturing		Impact	
The cutting index is not necessarily indicated if the performance letter is given (in that case, an X would replace the number)		From A to H An X indicates that the test has not been carried out (i.e. gloves that don't use the blade during the test)		Not indicated if not tested	

Which letter should I use?

For BYCN:
Index 5 = D or +

From 21/04/2019, performance letters must be identified on gloves. Today, some suppliers are already applying the standard (on the glove or on the label on the glove).

What should I do with old gloves?

Don't panic! Gloves that were on the market before 21 April 2019 may still be sold until 2023, and will still be valid, so you can continue to use them.

For more information, see PPE Guide Specifications



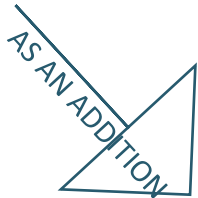
KRYTECH 586

Mechanical hazard: 4543



FOR WHAT TYPE OF ACTIVITY?

ASSEMBLING METAL FRAMEWORKS
WORKING IN GLOVES DURING COLD WEATHER
PACKING, LABELLING, WRITING
FORMWORK
ELECTRO-MECHANICAL/MECHANICAL WORK NOT AT VOLTAGE
FRAMEWORK
DELICATE MANIPULATION (ELECTRICIAN) (NOT AT VOLTAGE)
MANIPULATING PORTABLE ELECTRICAL EQUIPMENT
HANDLING HEAVY AND LIGHT ITEMS, RECONDITIONING ELECTRICAL EQUIPMENT, OTHER NON-LIVE EQUIPMENT
REPAIRING PORTABLE ELECTRICAL MACHINERY
WORK AT A GREAT HEIGHT (MANIPULATION)
NON-LIVE WORK
SUPERINTENDING, NON-OPERATIONAL JOBS / WORKSITE MANAGEMENT
WORK IN POLLUTED SOIL OR ON CONTAMINATED LAND



glove liners for working during cold weather



KELVAR OVERSLEEVES

Mechanical hazard: 132x

Thermal hazard: x1xxxx

(option in addition to gloves with mechanical strength)



KRYNIT 582

Mechanical hazard: 4543

Oily environment



FOR WHAT TYPE OF ACTIVITY?

OTHER NON-LIVE EQUIPMENT
FORMWORK
ELECTRO-MECHANICAL/MECHANICAL WORK NOT AT VOLTAGE
RECONDITIONING ELECTRICAL EQUIPMENT



FEELPRO

Mechanical hazard: 4121
Tactile



FOR WHAT TYPE OF ACTIVITY?

TRACING (AND ALSO TOPOGRAPHY)
VISITORS / SUPERINTENDING, NON-
OPERATIONAL JOBS / WORKSITE
MANAGEMENT

ULTRANE 551

Mechanical hazard: 4131
polyurethane foam



FOR WHAT TYPE OF ACTIVITY?

VISITORS
SUPERINTENDING, NON-OPERATIONAL
JOBS / WORKSITE MANAGEMENT

PICGUARD

Mechanical hazard: 4444



FOR WHAT TYPE OF ACTIVITY?

WORK IN POLLUTED SOIL OR ON
CONTAMINATED LAND
WORK IN INSANITARY CONDITIONS

LEATHER WELDER QUILTED LINING

Mechanical hazard: 4244
Thermal hazard: 444x4x



FOR WHAT TYPE OF ACTIVITY?

OPERATIONS NEAR HOT SPOTS (500
DEGREES FOR 15 SEC)
OXYGEN CUTTING
WELDING DEBURRING

TIG WELDER WITH 15 OVERSLEEVE

Mechanical hazard: 3122
Thermal hazard: 412x4x



FOR WHAT TYPE OF ACTIVITY?

OXYGEN CUTTING
WELDING DEBURRING



ALPHATEC WITH OVERSLEEVE

Mechanical hazard: 4121

Chemical hazard: J6K6L3

Leakproofing: NQA3



FOR WHAT TYPE OF ACTIVITY?

CONCRETING/MAJOR CONCRETE CONSTRUCTIONS

MANIPULATING PETROLEUM PRODUCTS AND SOLVENTS

PREPARING AND USING CHEMICALS

PAINTING / PAINTING IN BOOTHS

MAKING RESIN JUNCTION BOXES (CONNECTING TO NETWORK)

BLUE NEOPRENE FOR CEMENTING

Mechanical hazard: 4131

Chemical hazard: A2K6L4

Thermal hazard: x1xxxx



FOR WHAT TYPE OF ACTIVITY?

CONCRETING/MAJOR CONCRETE CONSTRUCTIONS

PREPARING AND USING CHEMICALS

PVC SUPER ACTIFRESH FOR CHEMICALS WITH 40 OVERSLEEVE

Mechanical hazard: 4121

Chemical hazard: A2K6L3



FOR WHAT TYPE OF ACTIVITY?

PREPARING AND USING CHEMICALS

PAINTING / PAINTING IN BOOTHS

MAKING RESIN JUNCTION BOXES (CONNECTING TO NETWORK)

PVC NEOPRENE FOR CHEMICALS

Mechanical hazard: 2111

Chemical hazard: A2K6L5

Leakproofing: NQA3



GREEN NITRIL FOR CHEMICALS

Mechanical hazard: 4101

Chemical hazard: A2J6K6L3





GREEN NITRIL FOR CHEMICALS WITH 18 OVERSLEEVE

Mechanical hazard: 4101
Chemical hazard: AJKL
Dexterity: 5



FOR WHAT TYPE OF ACTIVITY?

HOUSEWORK AND CLEANING
PREPARING AND USING PAINT /
PAINTING IN BOOTHS
CHEMICALS
MAKING RESIN JUNCTION BOXES
(CONNECTING TO NETWORK)

LATEX VITAL 124

Mechanical hazard: 1010
Dexterity: latex



FOR WHAT TYPE OF ACTIVITY?

ASBESTOS (AND CERAMIC FIBRE)
HANDLING AND LEAD
HOUSEWORK AND CLEANING



Over cut-resistant gloves

NATURAL RUBBER

Mechanical hazard: x010
Chemical hazard: NQA3



FOR WHAT TYPE OF ACTIVITY?

ASBESTOS (AND CERAMIC FIBRE)
HANDLING AND LEAD
HOUSEWORK AND CLEANING



Over cut-resistant gloves

POWERFIT

Mechanical hazard: 4343
Dexterity : 5



FOR WHAT TYPE OF ACTIVITY?

STRIPPING CABLE INSULATION (NOT LIVE)
FINISHING WORK; DELICATE HANDLING,
JUNCTIONS, ETC. (SUB-TRADES AND
RESTORATION WORK)
REPAIRING PORTABLE ELECTRIC
MACHINERY, SMALL STRIPPING JOBS
(DISTRIMO)



VEGA 27 for oily environments

Mechanical hazard: **4343**

Dexterity: 5

100% nitrile coating

Leakproof



LEBON oversleeve MANC/TNB/DC/P

Mechanical hazard: **4121**



FOR WHAT TYPE OF ACTIVITY?

FINISHING WORK: DELICATE HANDLING, JUNCTIONS, ETC. (STRUCTURAL WORK, SUB-TRADES AND RESTORATION WORK)
HIGH-PRESSURE WASHING, DESCALING
HEAVY DEMOLITION WORK

RIPEURCO LEATHER

Mechanical hazard: **4544**



FOR WHAT TYPE OF ACTIVITY?

BEATING
BRUSH CUTTING
HANDLING ITEMS
PACKING, LABELLING, WRITING
HEAVY DEMOLITION WORK
CHOPPING (EXCLUDING CHAINSAW USE), RIPPING, CUTTING (WITH SPARKING)

PRUNING GLOVE

Mechanical hazard: **3132**

EN381-7: Cat.1



FOR WHAT TYPE OF ACTIVITY?

PRUNING (CHAIN SAW)



PRUNING OVERSLEEVE

EN381-7: Cat.1

Protecting the hands



Insulating glove shall be used with a leather overglove if isn't composite. The date of the electrical test at the end of manufacture is shown on each glove. Gloves used frequently shall be tested at intervals of less than 6 months. Gloves used occasionally shall be tested after use and at least every 12 months. The slightest puncture renders the glove unusable. Before each use, inspect the glove visually and check it by inflating.

Electrical resistance - EN 60903: **AZC**
Size: **7 - 11**



INSULATING GLOVE TST 500V CL00

Work at a maximum operating voltage of 500 volts.
Electrical resistance - EN 60903: **AZC**
Size: **7 - 11**

INSULATING GLOVE TST 1000V CLO

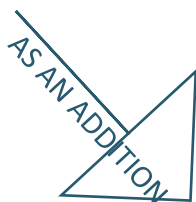
Work at a maximum operating voltage of 1000 volts.
Electrical resistance - EN 60903: **RC**
Size: **8 - 11**

INSULATING GLOVE TST 7500V CL1

Work at a maximum operating voltage of 7500 volts.
Size: **8 - 11**

INSULATING GLOVE TST 26500V CL3

Work at a maximum operating voltage of 26500 volts.
Electrical resistance - EN 60903: **RC**
Size: **9 - 11**



TST HV LEATHER OVERGLOVE



Protects Electrosoft gloves against cuts and tears.

Length 31.5cm. Thickness: Palm: 0.8/1.1 mm
- Oversleeve: 1.3/1.5 mm. Dexterity class 5.
Mechanical strength ((abrasion, cuts, tears, punctures) - EN 388: **2121**
Size: **8 - 11**

NEW !!!

INSULATING TST 500V CL00 COMPOSITE

Work at a maximum operating voltage of 1000 volts.
Electrical resistance - EN 60903: **RC**

No need for leather overgloves!

FOR WHAT TYPE OF ACTIVITY?

**CONNECTING/DISCONNECTING
NETWORKS**

**ELECTRICAL WORK: OPERATIONS ON
BATTERIES AT VOLTAGE, TAKING
MEASUREMENTS: WORK AT VOLTAGE:
LV AND HV (+ OVERGLOVES IF ISN'T
COMPOSITE)**

17





ESPUNA LEATHER

Mechanical hazard: 3122

Dexterity: 5



FOR WHAT TYPE OF ACTIVITY?

PACKING, LABELLING, WRITING
HANDLING ITEMS

PEAUCEROS LEATHER

Mechanical hazard: 2121

Dexterity: 5

Artery protection



VIBRAGUARD

Mechanical hazard: 3221

EN10819



FOR WHAT TYPE OF ACTIVITY?

MANIPULATING PORTABLE ELECTRICAL EQUIPMENT
OR VIBRATING NEEDLE

VIBRAPROTECT

Mechanical hazard: x243

>300hz

EN10819



WINTERSAFE

Mechanical hazard: 4542

EN511: 02x



FOR WHAT TYPE OF ACTIVITY?

WORKING DURING COLD WEATHER



DISPOSABLE VINYL gloves

EN374-5: bacteria, fungi



FOR WHAT TYPE OF ACTIVITY?
CARE

DISPOSABLE LATEX gloves

EN374-3



PVC SUPER ACTIFRESH CHEMICAL WITH 40 OVERSLEEVE

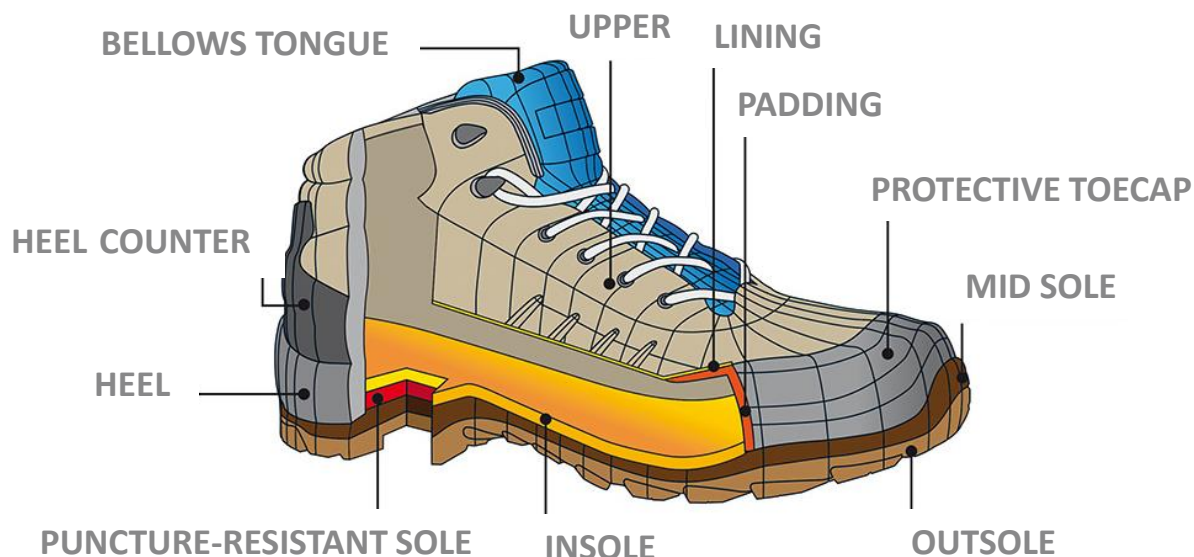
Mechanical hazard: 4121

EN374-3: A2K6L3



FOR WHAT TYPE OF ACTIVITY?
WORK IN INSANITARY CONDITIONS





MALLEOLUS PROTECTOR: top part of the upper or the part at the front,
Foam-lined to protect against impacts on the middle of the foot
and up to ankle height.

THREE TYPES OF SAFETY FOOTWEAR ARE THEREFORE DEFINED IN THIS GUIDE:

- High-top shoes (Ranger type)
- High-cut shoes (between low and high top)
- Low-cut shoes

*For more information,
see PPE Guide Specifications*



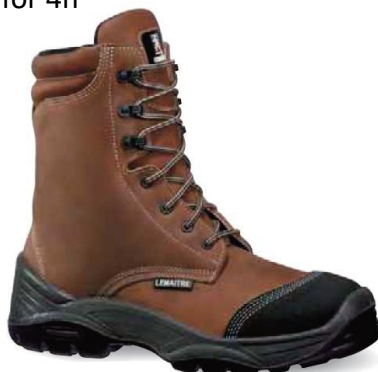


JORAN S3 CI SRC HIGH-TOP SHOE

leather water repellent for 4h

(EN 42: 695g)

Size : 35 to 48



JALSBORG HIGH-TOP SHOE S3 CI SRC

leather water
repellent for 8h

(EN 42: 750g)

Size : 35 to 48



JALOSBERN STEEL S3 CI SRC

leather water repellent
for 8h

(EN 42: 750g)

Size : 38 to 48

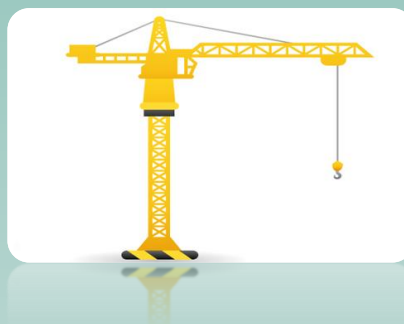


JALOSBERN STEEL Winter S3 CI SRC

leather water repellent
for 8h

(EN 42: 875g)

FOR WHAT TYPE OF ACTIVITY?



**GENERAL ACTIVITIES (STRUCTURAL
WORK, EXCLUDING SPECIAL
REQUIREMENTS)
SUPERINTENDING, NON-OPERATIONAL
JOBS / SUPERVISION / WORKSITE
MANAGEMENT**

FOR WHAT TYPE OF ACTIVITY?
WORK IN COLD CONDITIONS

SRA/SRB OR SRC = slip-resistant sole

leather water repellent : see the recommendations supplier



Overshoes with composite puncture-resistant sole and steel shell

Size : 35 to 48



FOR WHAT TYPE OF ACTIVITY?

MOVING AROUND AN INDUSTRIAL STORE OR WORKSHOP, OCCUPIED SITE (SALES STAFF, SUPERVISORS AND VISITORS ETC.)

OUTSIDE THE WORK SITE

High-cut DIABLO S3 CI SRC

water repellent leather
(EN 42: 700g)

Size : 35 to 48



FOR WHAT TYPE OF ACTIVITY?

DRIVERS OF MACHINERY

PREPARING PACKAGES, LOADING, UNLOADING, RECEPTION, STORE LOGISTICS, AFTER-SALES SERVICE REPAIRING PORTABLE ELECTRICAL MACHINERY AS AN AFTER-SALES SERVICE

FINISHING JOBS INVOLVING SUSTAINED KNEELING (E.G. FIXING SKIRTING BOARDS, LAYING PAVING, ETC.)

High-cut ATLANTIC OVERCAP S3 HI CI SRC

Water repellent leather
(EN 42: 900g)

Size : 38 to 48



High-cut ALBI S3 CI SRC

(EN 42: 650g)

Size : 38 to 48



SAV



SRA/SRB OR SRC = slip-resistant sole

leather water repellent : see the recommendations supplier

PPE Guide - 05/19

SPECIFIC



S5 SRA PVC boots

(650g)
Size : 35 to 48



FOR WHAT TYPE OF ACTIVITY?

CONCRETING
POURING CONCRETE, BUILDING WORK
WASHING (NOT AT PRESSURE)
OUTSIDE VISITORS TO THE WORK SITE

PU S5 SRC CI boots

(500g)
Size : 37 to 48



FOR WHAT TYPE OF ACTIVITY?

CONCRETING
POURING CONCRETE, BUILDING WORK
FOR CONTINUOUS WEAR

High-top CALGARY S3 SRC

0% metal
Size : 38 to 48



FOR WHAT TYPE OF ACTIVITY?

ALL BASIC T&D WORK AND TRAVELLING
OUTSIDE
ELECTRICAL WORK
WORK IN THE MOUNTAINS, PYLONS



High-cut DUNDEE S3 SRC

0% metal
(EN 42: 650g)
Size : 39 to 47



FOR WHAT TYPE OF ACTIVITY?

ALL BASIC T&D WORK AND TRAVELLING
OUTSIDE
ELECTRICAL WORK
WORK IN THE MOUNTAINS, PYLONS
AIRPORT ENVIRONMENTS



SPECIFIC



Boots PRIMO 800

(EN 42: 715g)
Size : 40 to 48



FOR WHAT TYPE OF ACTIVITY?
HP WASHING UP TO 800 BARS

Disposable overshoes



FOR WHAT TYPE OF ACTIVITY?
MOVING IN THE WHITE ROOM

High-top ARGONO S3 HRO SRC

Size : 38 to 48



FOR WHAT TYPE OF ACTIVITY?
WELDER BOILERMAKER

Leather GAITER protecting the shoe



GAITER for chain sawing

EN381-9: Cat.1

Size : 39 to 46



FOR WHAT TYPE OF ACTIVITY?
CHAIN SAWING





Low-cut ALES S3 CI SRC

(en 42 : 530g)

Size : 38 to 48



Low-cut DUCIE S3 SRC 0% métal

(en 42 : 530g)

Size : 39 to 47



Low-cut SWIM S1P SRC 0% métal

Size : 39 to 47



FOR WHAT TYPE OF ACTIVITY?

FINISHING JOBS INVOLVING SUSTAINED KNEELING (E.G. FIXING SKIRTING BOARDS, LAYING PAVING, LAYING FLOORING ETC.)
PREPARING PACKAGES, LOADING, UNLOADING, RECEPTION, STORE LOGISTICS, AFTER-SALES SERVICE
REPAIRING PORTABLE ELECTRICAL MACHINERY AS AN AFTER-SALES SERVICE
AIRPORT ENVIRONMENTS

> **VALIDATION BY ENTITY**

**High-cut
and high-top shoes
in priority**





STANDARDS / RECOMMENDATIONS

Marking on the frame

The marking on the frame must include the EC marking, the manufacturer's identification (brand or logo), the number of the EN standard, the protection symbols indicating mechanical strength and the fields where it may be used.

THE DIFFERENT PROTECTION SYMBOLS

■ FOR MECHANICAL STRENGTH

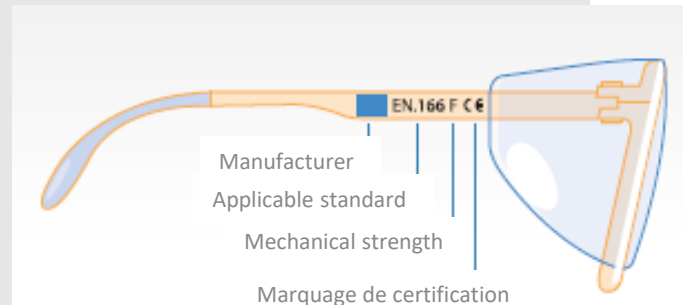
- ★ **A:** high-energy impact (strength test with a 6mm sphere weighing 0.86g at 190 m/s)
- ★ **B:** medium-energy impact (strength test with a 6 mm sphere weighing 0.86g at 120 m/s)
- ★ **C:** low-energy impact (strength test with a 6 mm sphere weighing 0.86g at 45 m/s)
- ★ **S:** increased robustness (strength test with a 22 mm sphere weighing 43g falling from a height of 1.30m)

If the frame has no symbol, it is classed as having minimum robustness

■ FOR FIELDS OF USE

- ★ **3:** liquid droplets or splashes
- ★ **4:** coarse dust particles > 5 microns
- ★ **5:** gas and fine dust < 5 microns
- ★ **8:** electrical arc from a short-circuit
- ★ **9:** molten metal and hot solids

If the frame has no symbol, it is classed as for general-purpose use



Marking on eyepieces

The marking on eyepieces must include the manufacturer's identification (brand or logo), the various symbols indicating mechanical strength, the optical class and the scale numbers (indicating the shade) for filtering eyepieces.

SYMBOLS FOR OPTICAL CLASS	
1	Continuous work
2	Intermittent work
3	Occasional work may not be worn all the time
9	Molten metal and hot solids

IMPORTANT

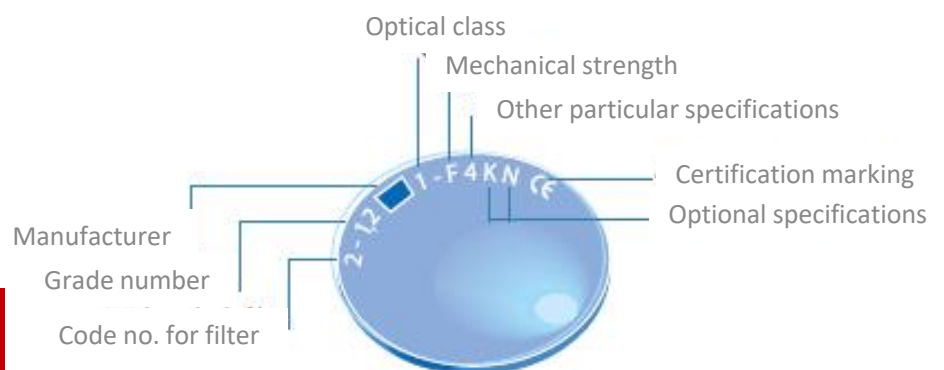
The symbols F, B and A give the maximum protection index for:

F: max protection for safety glasses

B: max protection for safety goggles

A: max protection for face protectors

If the symbols S, F, B and A are not common to the eyepiece and the frame, the lowest level shall be assigned to the entire protector.



For more information,
see PPE Guide Specifications



FOR WHAT TYPE OF ACTIVITY? ALL ACTIVITIES

RUSH+

Wide field of vision
2 sizes (*small or classic*),
Tinted and clear,
May be worn all the time,
Plastic frame
FT impact rating,
Scratch resistant,
Mist resistant
(26g)



CONTOUR

Tinted, half-tinted and clear,
May be worn all the time,
Plastic frame
FT impact rating,
Scratch resistant,
Mist resistant
(21g)



CAUTION: Tinted glasses for outdoors only. If interior and exterior work, provide semi tinted or have 2



SOLIS SOLIPOL

Polarizing,
May be worn all the time
Plastic frame
FT impact rating,
Scratch resistant,
(25g)



FOR WHAT TYPE OF ACTIVITY?
DRIVERS OF MACHINERY



POLYCARBONATE OVERGLASSES

Clear,
May be worn all the time
F impact rating,
UV2 filter
(70g)



FOR WHAT TYPE OF ACTIVITY?
OVERGLASSES OR VISITOR GLASSES



TRACKER

Tinted and clear,
May be worn all the time,
Plastic frame,
Foam reinforced,
FT impact rating,
Scratch resistant,
Mist resistant
(52g)



FOR WHAT TYPE OF ACTIVITY?

WINDY, SANDY OR DUSTY ENVIRONMENTS

BAXTER

Tinted and clear,
May be worn all the time,
Plastic frame,
Foam reinforced,
FT impact rating,
Scratch resistant,
Mist resistant
(49g)
Newer model of the Tracker





ULTRASONIC GOGGLES

May be worn all the time,
Clear,
BT impact rating,
UV2 filter,
Mist resistant,
Scratch resistant,
Field of use 9
(80g)



FOR WHAT TYPE OF ACTIVITY?

**GRINDING, CHISELLING
SANDBLASTING, CUTTING, SANDING
PAINTING WORK
BREAKING APART - FLANGE GASKETS
CLEANING WORK (DEGREASING)
SPASHES FROM HOT LIQUIDS, SPARKS,
GASES, FUMES, AEROSOLS AND NON-
HAZARDOUS LIQUIDS**

BLAST GOGGLES

May be worn all the time,
Clear,
BT impact rating,
UV2 filter,
Mist resistant,
Field of use 9
(113g)



BLAST FACE SHIELD

Clear,
BT impact rating,
Field of use 3
(80g)



(Option for Blast goggles)

FOR WHAT TYPE OF ACTIVITY?

SANDBLASTING, CUTTING, SANDING

ARCO SHIELD

Compatible with
IRIS/KARA/BRENNUS helmets,
Clear,
B impact rating,
Field of use 3,
(115g)





CHAMELEON WELDER MASK

Shades 4 - 13



FOR WHAT TYPE OF ACTIVITY?

WELDING

COVERSAL FLIP-UP SHOT-BLASTING GOGGLES

Shade filter 5,
F impact rating,
May be worn all the time,
Interchangeable eyepieces,
Compatible with ordinary specta
(129g)



WELDING GOGGLES WITH SAFETY HELMET WUITHOM Electronique

Shades 8 - 14
(980g)



FOR WHAT TYPE OF ACTIVITY?
TORCH WELDING





BUBBLE SHIELD FOR IDRA 2 HELMET

UV2 filter

(Note: cannot be adapted to noise-resistant shells)



FOR WHAT TYPE OF ACTIVITY?

**ELECTRICAL DISCONNECTION WHEN LIVE
ELECTRICAL (ELECTRIC ARC - MOLTEN
METAL)
RADIATION FROM ELECTRICAL ARCING
CONNECTION WORK**



ARCO SHIELD

Compatible with
IRIS/KARA/BRENNUS helmets,
Clear,
B impact rating,
Field of use 3,
(115g)





MOULDED EAR PLUGS



(see BY Purchasing have framework contract)

FOR WHAT TYPE OF ACTIVITY?

WORN ALL THE TIME (FOLLOWING THE RISK ASSESSMENT)

ALL ACTIVITIES

MACHINERY REPAIR; OPERATIONAL TEST (ACOUSTIC PANELS ABOVE THE WORKSHOP FOR STAFF IN THE VICINITY)

HIGH-PRESSURE WASHING AT 400 BAR

SPECIFIC

EAR PROTECTORS FOR IRIS/KARA HELMET

Metal
31dB
(317g)



FOR WHAT TYPE OF ACTIVITY?

HIGH-PRESSURE WASHING AT 400 BAR

JOB IN A NOISY ENVIRONMENT (+++)

(E.G. AIRPORTS AND DEMOLITION SITES)

NOISE-RESISTANT HELMET EAR MUFFS

Metal
28dB
(194g)



NOISE-RESISTANT HELMET C3 EAR MUFFS

Plastic
33dB
(302g)



NOISE-RESISTANT HELMET NECKBAND

35dB
(270g)



EAR PROTECTORS FOR C1H HELMET

0% metal
26dB
(280g)
+ moulded ear plugs





DISPOSABLE MOULDED EAR PLUGS FOR A DISPENSER

33dB



FOR WHAT TYPE OF ACTIVITY?

EAR PLUGS FOR VISITORS / OCCASIONAL NEEDS

EAR-PLUG DISPENSER

REUSABLE EAR PLUGS WITH CORD IN CASE

30dB
(3.9g)



DISPOSABLE EAR PLUGS WITH CORD

37dB
(1g)



BOUCHONS FIT IN



Attenuation 26dB.
Available in 21dB and 23dB also.

For who ?

Occasional, Interim and Subcontractor Populations

This does not replace the Moulded ear plugs of our employees.



STANDARDS / RECOMMENDATIONS

THE VARIOUS AIR POLLUTANTS

- **DUST AND SMOKE:** during grinding, sand blasting, sanding and crushing materials, particles of different sizes become suspended in the air.
- **MISTS:** when products are sprayed or condensed, small droplets of liquid remain suspended in the air.
- **VAPOURS:** substances in the gaseous state (liquids or solids), formed when the liquids or solids evaporate (e.g. solvents and hydrocarbons)
- **GASES:** such as air. They may be simple or complex, and spread readily at ambient temperatures. Some are very hazardous and require special protection..

■ TWA: TIME-WEIGHTED AVERAGE EXPOSURE

Average concentration for an 8-hour working day and a 40-hour working week. This daily exposure represents no health risk.

■ STEL: SHORT-TERM EXPOSURE LIMIT

Concentration below which individuals may be exposed for 15 minutes without risk to their health.



*For more information,
see PPE Guide Specifications*



Use in degraded mode: when local exhaust ventilation is not possible

<15mn

FFP3V DISPOSABLE HALF-FACE MASK



FOR WHAT TYPE OF ACTIVITY?

GENERAL

TAPPING, SANDING, SWEEPING, CUTTING
WOOD, GRINDING
WORK IN AN ENVIRONMENT WITH
CEMENT OR SILICA DUST
CLEANING MACHINERY (AFTER-SALES
SERVICE)

FFA2P3D DISPOSABLE HALF-FACE MASK

FOR WHAT TYPE OF ACTIVITY?

PREPARATION BEFORE PAINTING
PAINTING
PAINTING WORK OUTSIDE BOOTHS
PAINTING WORK IN BOOTHS
WORK IN CONFINED AREAS
USE OF CHEMICALS HYDROCARBON
VAPOUR

FFABEK1P3D DISPOSABLE HALF-FACE MASK



Indication
de
protection



Use of chemicals: also depends on the recommendations in the product's Safety Data Sheet
Asbestos / lead = special case where the dust level generated by the particular operating procedure determines which PPE is suitable. Please contact the Risk Prevention Department





<1H

FORCE 8 HALF-FACE MASK WITH PRESS-TO-CHECK



FOR WHAT TYPE OF ACTIVITY?

GENERAL

CLEANING MACHINERY (AFTER-SALES SERVICE)

TAPPING, SANDING, SWEEPING, CUTTING WOOD, GRINDING

PREPARATION BEFORE PAINTING

PAINTING

WORK IN AN ENVIRONMENT WITH CEMENT OR SILICA DUST

ELECTRICAL WORK ON AIR COOLERS

WORK ON ELECTRICAL EQUIPMENT

WITH CERAMIC FIBRE PRESENT

PAINTING WORK OUTSIDE BOOTHS

PAINTING WORK IN BOOTHS

WORK IN CONFINED AREAS

USE OF CHEMICALS HYDROCARBON VAPOUR

P3R PRESS-TO-CHECK FILTER

FOR WHAT TYPE OF ACTIVITY?

GENERAL

CLEANING MACHINERY (AFTER-SALES SERVICE)

TAPPING, SANDING, SWEEPING, CUTTING WOOD, GRINDING

WORK IN AN ENVIRONMENT WITH CEMENT OR SILICA DUST



A2P3R PRESS-TO-CHECK FILTER



FOR WHAT TYPE OF ACTIVITY?

ELECTRICAL WORK ON AIR COOLERS

WORK ON ELECTRICAL EQUIPMENT WITH CERAMIC FIBRE PRESENT

HYDROCARBON VAPOURS

ABEK1P3R PRESS-TO-CHECK FILTER



FOR WHAT TYPE OF ACTIVITY?

PREPARATION BEFORE PAINTING

PAINTING

PAINTING WORK OUTSIDE BOOTHS

PAINTING WORK IN BOOTHS

WORK IN CONFINED AREAS

USE OF CHEMICALS





9000 SERIES FULL-FACE MASK (370g)



FOR WHAT TYPE OF ACTIVITY?

GENERAL

TAPPING, SANDING, SWEEPING,

CUTTING WOOD, GRINDING

PREPARATION BEFORE PAINTING

PAINTING

WORK IN AN ENVIRONMENT WITH

CEMENT OR SILICA DUST

PAINTING WORK OUTSIDE BOOTHS

PAINTING WORK IN BOOTHS

WORK IN CONFINED AREAS

USE OF CHEMICALS

P3R FILTER FOR 9000 SERIES (30g)



FOR WHAT TYPE OF ACTIVITY?

GENERAL

TAPPING, SANDING, SWEEPING, CUTTING

WOOD, GRINDING

WORK IN AN ENVIRONMENT WITH

CEMENT OR SILICA DUST

ABEK FILTER FOR 9000 SERIES (92g)



FOR WHAT TYPE OF ACTIVITY?

PAINTING

PAINTING WORK OUTSIDE BOOTHS

PAINTING WORK IN BOOTHS

PREPARATION BEFORE PAINTING

WORK IN CONFINED AREAS

USE OF CHEMICALS

Contact distrimo or prevention service for filter adapted to the task



POWERFLOW FULL-FACE MASK



P3R FILTER FOR POWERFLOW



FOR WHAT TYPE OF ACTIVITY?

TORCH-WELDING LEADED METAL ITEMS

GENERAL

TAPPING, SANDING, SWEEPING,

CUTTING WOOD, GRINDING

PREPARATION BEFORE PAINTING

PAINTING

WORK IN AN ENVIRONMENT WITH

CEMENT OR SILICA DUST

PAINTING WORK OUTSIDE BOOTHS

PAINTING WORK IN BOOTHS

WORK IN CONFINED AREAS

DE-SCALING WORK, LEAD-BASED

PAINTING

USE OF CHEMICALS

Contact distrimo or prevention service for filter adapted to the task





CLEANSPACE 2 HALF-FACE MASK WITH ASSISTED VENTILATION



FOR WHAT TYPE OF ACTIVITY?

GENERAL

TAPPING, SANDING, SWEEPING, CUTTING
WOOD, GRINDING

WORK IN AN ENVIRONMENT WITH
CEMENT OR SILICA DUST

CLEANSPACE 2 FACE MASK WITH ASSISTED VENT



FOR WHAT TYPE OF ACTIVITY?

GENERAL

TAPPING, SANDING, SWEEPING, CUTTING
WOOD, GRINDING

WORK IN AN ENVIRONMENT WITH
CEMENT OR SILICA DUST
SHOT-BLASTING IN BOOTHS

A1P3R FILTER FOR CLEANSPACE 2





CLEANSPEACE MASK WITH HIGH-PERFORMANCE ASSISTED VENTILATION



FOR WHAT TYPE OF ACTIVITY?
ASBESTOS/LEAD



P3R FILTER FOR CLEANSPEACE ULTRA



Contact distrimo or prevention service for filter adapted to the task



>1H

VERSAFLO MASK WITH ASSISTED VENTILATION



FOR WHAT TYPE OF ACTIVITY?

GENERAL

TAPPING, SANDING, SWEEPING, CUTTING
WOOD, GRINDING

WORK IN AN ENVIRONMENT WITH
CEMENT OR SILICA DUST
SHOT-BLASTING IN BOOTHS

VERSAFLO WELDER MASK WITH ASSISTED VENTILATION



FOR WHAT TYPE OF ACTIVITY?

TORCH WELDING/WELDING
EVACUATION
LABORATORY

P3R FILTER FOR VERSAFLO



FOR WHAT TYPE OF ACTIVITY?

GENERAL

TAPPING, SANDING, SWEEPING, CUTTING
WOOD, GRINDING

WORK IN AN ENVIRONMENT WITH
CEMENT OR SILICA DUST
SHOT-BLASTING IN BOOTHS

ABEKP3R FILTER FOR VERSAFLO

FOR WHAT TYPE OF ACTIVITY?

TORCH WELDING/WELDING
EVACUATION
LABORATORY





COMPACT AIR VENTILATED BELT



FOR WHAT TYPE OF ACTIVITY?

TORCH-WELDING LEADED METAL ITEMS
GENERAL
TAPPING, SANDING, SWEEPING,
CUTTING WOOD, GRINDING
WORK IN AN ENVIRONMENT WITH
CEMENT OR SILICA DUST
DE-SCALING WORK, LEAD-BASED
PAINTING

HALF MASK COMPACT AIR



P3

FILTER FOR COMPACT AIR



POWERCAP
VENTILATED
TH1



Turbolite 2m automatic fall-arrestor



FOR WHAT TYPE OF ACTIVITY?

CRANE ERECTOR
ERECTING HUTS (WITH ANCHORING ON HUT)
METAL FRAMEWORK ERECTOR
WORKING AT HEIGHT

STANDARDS / RECOMMENDATIONS

Equipment and accessories to be defined depending on the work situation (cradle, unloading bay, flat roof, etc.).

Contact your Risk-Prevention Department.

To use a harness, I must be trained and authorized.

Guide to sizes

	S	M	XL	
A	60 - 100	80 - 120	100 - 160	
B	75 - 100	90 - 110	100 - 125	
C	45 - 80	45 - 100	45 - 120	
D	55 - 65	65 - 80	80 - 90	

For more information,
see PPE Guide Specifications





TRACTEL HT55BY HARNESS

HONEYWELL RM HARNESS



TRACTEL HT ELECTRA HARNESS

HONEYWELL RM HARNESS

TRACTEL HT55BY HARNESS



RUNNER 150cm

IK 208 2m positioning lanyard

Fall-arrest lanyard with 1.5m

Manyard ME68 shock absorber



1.80m automatic fall-arrestor with shock absorber



10m automatic fall-arrestor
RK1 karabiner with automatic locking

FOR WHAT TYPE OF ACTIVITY?

RAISING EQUIPMENT TO A HEIGHT
ERECTING HUTS (WITH ANCHORING ON HUT)
WORKING AT HEIGHT
WORKING ON A CRADLE

FOR WHAT TYPE OF ACTIVITY?

EHV PYLON ERECTORS
WORKING AT HEIGHT

FOR WHAT TYPE OF ACTIVITY?

HEAVY-GOODS DRIVER
ELECTRICAL INSTALLATION
FORMWORK ERECTOR
CRANE ERECTOR
ERECTING HUTS (WITH ANCHORING ON HUT)
METAL FRAMEWORK ERECTOR
WORKING AT HEIGHT

FOR WHAT TYPE OF ACTIVITY?

HEAVY-GOODS DRIVER
ELECTRICAL INSTALLATION
METAL FRAMEWORK ERECTOR
WORKING AT HEIGHT

FOR WHAT TYPE OF ACTIVITY?

HEAVY-GOODS DRIVER
ELECTRICAL INSTALLATION
METAL FRAMEWORK ERECTOR
ERECTING CRANES
WORKING AT HEIGHT





FOR WHAT TYPE OF ACTIVITY?

**WORKSHOP, MECHANICAL
WELDING, TORCH WELDING**

TABLIER DE SOUDEUR 90cm ou 110cm	VESTE DE SOUDEUR	GUETRES DE SOUDEUR	CAGOULE DE SOUDEUR	GANTS DE SOUDEUR	LUNETTES SOUDEUR	MASQUE SOUDEUR
-------------------------------------	------------------	---------------------------	--------------------	------------------	------------------	----------------

Presented previously in this Guide



FOR WHAT TYPE OF ACTIVITY?

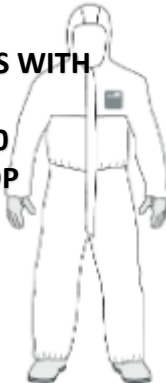
CHOPPING WOOD

CUT-RESISTANT LEGGINGS



LEGGINGS E

**DISPOSABLE TYVEK COVERALLS WITH THUMB LOOP
DISPOSABLE MICROGARD 2000
COVERALLS WITH THUMB LOOP**



FOR WHAT TYPE OF ACTIVITY?

**PAINTING IN BOOTHS
ASBESTOS (TYVEK)**

COMBINAISON JETABLE 3M BLANCHE	COMBINAISON JETABLE 3M BLEU / BLANCHE	COMBINAISON JETABLE TYPE 5 BLANCHE	COMBINAISON JETABLE TYPE 4-5-6 BLANCHE AVEC RABAT AUTOCOLLANT	COMBINAISON JETABLE TYVEK TYPE 5-6 BLANCHE	COMBINAISON JETABLE TYVEK TYPE 5-6 BLEUE
-----------------------------------	--	---------------------------------------	---	---	---



FOR WHAT TYPE OF ACTIVITY?

LABORATORY, INSANITARY CONDITIONS





GILET DE SAUVETAGE MANUEL PILOT 165	GILET DE SAUVETAGE AUTOMATIQUE PILOT 165	BOUEE COURONNE D.Ext 75cm
--	---	----------------------------------

FOR WHAT TYPE OF ACTIVITY?

PROXIMITY TO WATER

MARITIME WORK



FOR WHAT TYPE OF ACTIVITY?

CONFINED SPACES

DETECTEUR MONOGAZ O2, CO, CO2 ou H2S	DETECTEUR MULTIGAZ ALTAIR 4X O2 + H2S + CO + GAZ EXPLO	DEMI MASQUE D'EVACUATION MINISCAPE ABEK (5min)	DEMI MASQUE D'EVACUATION ABEK (15min)
---	---	---	--



FOR WHAT TYPE OF ACTIVITY?

ASBESTOS/LEAD

COMBINAISON JETABLE TYVEK BLANCHE OU BLEUE	ROULEAU DE SCOTCH TOILE AMERICAINE OU ORANGE	SUR BOTTES ANTI DERAPANTE	LINGETTES NETTOYANTES	MASQUE COMPLET POWERFLOW	MASQUE CLEANSAPCE ULTRA VENTILATION ASSISTEE (pour amiante/plomb)	GANTS LATEX NATUREL
---	---	----------------------------------	------------------------------	---------------------------------	---	----------------------------



Presented previously in this Guide

FOR WHAT TYPE OF ACTIVITY? UNCOMFORTABLE TEMPERATURES

KIT HIVER

FLEECE UNDER-HELMET BONNET + HV NECK
PROTECTOR + MUFFLER
WATER-REPELLENT UNDER-HELMET HOOD
COLD-RESISTANT GLOVE LINERS
WINTERSAFE COLD-RESISTANT GLOVES
IC 5
NON-SLIP SOLES
LINER FOR COSMOS BOOTS
FLEECE NECK WARMER
SOCKS
PROTECTIVE COLD-WEATHER BONNET



KIT ETE

SUMMER COOLING COWL FOR CROWN OF HEAD
AND NECK
MESH WAISTCOAT
SUMMER WAISTCOAT
SUN CAPE NAPE PROTECTOR ORANGE UV
ABSORBANT



PPE SHEET

I am a visitor from outside

EYE PROTECTION (p.34 35)



HEAD PROTECTION (p.8)

Model of helmet defined by the subsidiary



HEARING PROTECTION (p.40 - 41)

disposable plugs



HAND PROTECTION (p.16 17)



WORK CLOTHING

High visibility clothing (parka, jacket or waistcoat)



SAFETY BOOTS (p.26 28)



Note, this sheet is provided for information only, and is not intended to be comprehensive. Personal protection shall be adapted to each work situation after a risk analysis

PPE SHEET

I am an employee

EYE PROTECTION (p.34)

Safety eyewear is available



HEAD PROTECTION (p.8)

Model of helmet defined by the subsidiary



HEARING PROTECTION (p.40)

Moulded ear plugs, otherwise disposable plugs



HAND PROTECTION (p.16-17)



WORK CLOTHING

High visibility clothing (parka, jacket or waistcoat)



SAFETY SHOES (p.26)



Note, this sheet is provided for information only, and is not intended to be comprehensive. Personal protection shall be adapted to each work situation after a risk analysis

PPE SHEET

I am digging / sanding / sawing

EYE PROTECTION (p.36)

Safety eyewear is available



RESPIRATORY PROTECTION (p.45 - 52)

Mandatory if there is no local exhaust ventilation



HAND PROTECTION (p.16 23)



HEARING PROTECTION (p.40)

Moulded ear plugs, otherwise disposable plugs + helmet



In add for digging

WORK CLOTHING

High visibility clothing (overalls or waistcoat)

SAFETY SHOES (p.26)



Note, this sheet is provided for information only, and is not intended to be comprehensive. Personal protection shall be adapted to each work situation after a risk analysis

PPE SHEET

I am concreting

EYE PROTECTION (p.34)

Safety eyewear is available



HEAD PROTECTION (p.8)

Model of helmet defined by the subsidiary



HEARING PROTECTION (p.40)

Moulded ear plugs, otherwise disposable plugs



HAND PROTECTION (p.18)



WORK CLOTHING

High visibility clothing (overalls or waistcoat)



SAFETY BOOTS (p.28)



Note, this sheet is provided for information only, and is not intended to be comprehensive. Personal protection shall be adapted to each work situation after a risk analysis

PPE SHEET



Asbestos/Lead

EYE PROTECTION (p.34 35)
Safety eyewear is available



HEAD PROTECTION (p.8)
Model of helmet defined by the subsidiary



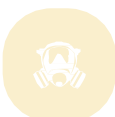
HEARING PROTECTION (p.40)
Moulded ear plugs, otherwise disposable plugs



RESPIRATORY PROTECTION (p.48 52 (lead), both p.50)



POWERFLOW 3M full-face mask



HAND PROTECTION (p.19)
Special gloves



WORK CLOTHING (p. 55 56)
Disposable coveralls



Blue coveralls

SAFETY SHOES (p.26)



Note, this sheet is provided for information only, and is not intended to be comprehensive. Personal protection shall be adapted to each work situation after a risk analysis

PPE SHEET

Chemicals

EYE PROTECTION (p.34 35)
Safety eyewear is available



HEAD PROTECTION (p.8)
Model of helmet defined by the subsidiary



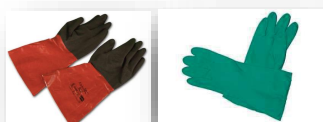
HEARING PROTECTION (p.40)
Moulded ear plugs, otherwise disposable plugs



RESPIRATORY PROTECTION (p.46 47 48)



HAND PROTECTION (p.18 19 23)



WORK CLOTHING (p.55 56)
High visibility clothing (overalls or waistcoat)



SAFETY SHOES (p.26)



Note, this sheet is provided for information only, and is not intended to be comprehensive.
Personal protection shall be adapted to each work situation after a risk analysis

PPE SHEET

I am an electrician

EYE PROTECTION (p.38)

Safety eyewear is available



HEAD PROTECTION (p.9)

Model of helmet defined by the subsidiary



HEARING PROTECTION (p.40)

Moulded ear plugs, otherwise disposable plugs



HAND PROTECTION (p.21)



WORK CLOTHING

High visibility clothing (overalls or waistcoat)/ arcflash)



SAFETY SHOES (p.28)



Note, this sheet is provided for information only, and is not intended to be comprehensive. Personal protection shall be adapted to each work situation after a risk analysis

Pablo codes

PPE	CODE
HEAD PROTECTION	
IRIS II HELMET	Avec PB + LUN + aération : EP008179+couleur+entité Avec PB +LUN sans aération : EP008180+couleur+entité Avec PB sans LUN + aération : EP008181+couleur+entité Avec PE + LUN + aération : EP008182+couleur+entité
KARA HELMET	Avec LUN + visière courte : EP008186+couleur+entité Avec LUN + visière standard : EP008187+couleur+entité Sans LUN + visière standard : EP008190+couleur+entité
BRENNUS HELMET	EP008175+couleur+entité
IDRA 2	EP008178+entité+couleur
E-MAN	EP004972
ALVEO BEST	EP004969
CHIN STRAP WITH 4 ATTACHMENT POINTS	EP002497
FIRST Base+	EP008191+couleur+entité
FIRST Base 3 XTRA	EP010629ORNON
Leather hood	EP006129
HAND PROTECTION	
KRYTECH 586	EP008084+taille (T7 à T11)
glove liners for working during cold weather	EP005110 ou EP005111
KELVAR OVERSLEEVES	EP002780
KRYNIT 582	EP008417+taille (T9 à T11)
FEELPRO	Contact distrimo
ULTRANE 551	EP008088+taille (T7 à T10)
PICGUARD glove liner	Sur commande
LEATHER WELDER QUILTED LINING	EP002333
TIG WELDER WITH 15 OVERSLEEVE	EP002334

PPE	CODE
ALPHATEC WITH OVERSLEEVE	EP008243+taille (T9 à T10)
BLUE NEOPRENE FOR CEMENTING	EP005113T10
PVC SUPER ACTIFRESH FOR CHEMICALS WITH 40 OVERSLEEVE	EP005680T10
PVC NEOPRENE FOR CHEMICALS	EP008246+taille (T9 à T10)
GREEN NITRIL FOR CHEMICALS	EP008247+taille (T9 à T10)
GREEN NITRIL FOR CHEMICALS WITH 18 OVERSLEEVE	EP005112T10
LATEX VITAL 124	EP009031+taille (T6 à T10)
NATURAL RUBBER	EP008245+taille (T9 à T10)
POWERFIT	EP005539+taille (T6 à T12)
VEGA 27 for oily environments	EP010344+taille (T7 à T10)
RIPEURCO LEATHER	EP008249+taille (T9 à T11)
PRUNING GLOVE	EP010568+taille (T8 à T12)
PRUNING OVERSLEEVE	EP010569
INSULATING GLOVE TST 500V CL00	EP008081+taille (T7 à T11)
INSULATING GLOVE TST 1000V CL0	EP008079+taille (T8 à T11)
INSULATING GLOVE TST 7500V CL1	EP008082+taille (T8 à T11)
INSULATING GLOVE TST 26500V CL3	EP008080+taille (T9 à T11)
TST HV LEATHER OVERGLOVE	EP008149+taille (T8 à T11)
INSULATING TST 500V CL00 COMPOSITE	EP008416+taille (T7 à T11)
ESPUNA LEATHER	EP008248+taille (T8 à T11)
PEAUCEROS LEATHER	EP005555+taille (T8 à T11)
VIBRAGUARD	EP008085+taille (T8 à T10)
VIBRAPROTECT	EP005536+taille (T8 à T10)
WINTERSAFE	EP010576+taille (T8 à T11)
DISPOSABLE VINYL gloves	EP005123T9
DISPOSABLE LATEX gloves	EP009216+taille (T6 à T9)

Pablo codes

PPE	CODE
FOOT PROTECTION	
JORAN S3 CI SRC HIGH-CUT SHOE	EP008228+pointure (35 à 48)
JALSBORG HIGH-CUT SHOE S3 CI SRC	EP017131+pointure (35 à 48)
JALOSBERN STEEL S3 CI SRC	EP008226+pointure (35 à 48)
JALOSBERN STEEL Winter S3 CI SRC	EP008390+pointure (35 à 48)
Overshoes with composite puncture-resistant sole and steel shell	EP005548N+pointure (35/39 à 45/48)
High-top DIABLO S3 CI SRC	EP008214+pointure (35 à 48)
High-top ATLANTIC OVERCAP S3 HI CI SRC	EP008429+pointure (38 à 48)
High-top ALBI S3 CI SRC	EP009803+pointure (38 à 48)
S5 SRA PVC boots	EP005544+pointure (37 à 48)
PU S5 SRC CI boots	EP005543+pointure (38 à 47)
High-cut CALGARY S3 SRC	EP008428+pointure (38 à 48)
High-top DUNDEE S3 SRC	EP008215+pointure (39 à 47)
Boots PRIMO 800	EP005541+pointure (39 à 48)
Disposable overshoes	EP007042 EP007043 (antidérapante)
High-top ARGONO S3 HRO SRC	EP008778+pointure (38 à 48)
Leather GAITER protecting the shoe	EP002401
GAITER for chain sawing	EP010570
EYE PROTECTION	
RUSH+	EP017886+teinte (IN, TE)
CONTOUR	EP008108+teinte (IN, ST, TE)
SOLIS SOLIPOL	EP008029PO
POLYCARBONATE OVERGLASSES	EP007910
TRACKER	EP008107+teinte (IN, TE)

PPE	CODE
BAXTER	À venir
ULTRASONIC GOGGLES	EP002730
BLAST GOGGLES	EP005755
BLAST FACE SHIELD	EP005756
ARCO SHIELD	EP002004
CHAMELEON WELDER MASK	EP002851
COVERSAL FLIP-UP SHOT-BLASTING GOGGLES	EP005761
WELDING GOGGLES WITH SAFETY HELMET WUITHOM Electronique	EP008463
BUBBLE SHIELD FOR IDRA 2 HELMET	EP002007
HEARING PROTECTION	
MOULDED EAR PLUGS	
EAR PROTECTORS FOR IRIS/KARA HELMET	EP001452
NOISE-RESISTANT HELMET EAR MUFFS	EP000496
NOISE-RESISTANT HELMET C3 EAR MUFFS	EP001454
NOISE-RESISTANT HELMET NECKBAND	EP007989
EAR PROTECTORS FOR C1H HELMET	EP001456
DISPOSABLE MOULDED EAR PLUGS FOR A DISPENSER	EP000266
EAR-PLUG DISPENSER	EP001879
REUSABLE EAR PLUGS WITH CORD IN CASE	EP000264
DISPOSABLE EAR PLUGS WITH CORD	EP000263

Pablo codes

PPE	CODE
RESPIRATORY PROTECTION	
FFP3V DISPOSABLE HALF-FACE MASK	EP005227
FFA2P3D DISPOSABLE HALF-FACE MASK	EP007985
FFABEK1P3D DISPOSABLE HALF-FACE MASK	EP007986
FORCE 8 HALF-FACE MASK WITH PRESS-TO-CHECK	EP010627
P3R PRESS-TO-CHECK FILTER	EP010628
A2P3R PRESS-TO-CHECK FILTER	EP010659
ABEK1P3R PRESS-TO-CHECK FILTER	EP010660
9000 SERIES FULL-FACE MASK (370g)	EP002850
P3R FILTER FOR 9000 SERIES (30g)	EP005086
ABEK FILTER FOR 9000 SERIES (92g)	EP005085
POWERFLOW FULL-FACE MASK	EP007988
P3R FILTER FOR POWERFLOW	EP007987
CLEANSFLO 2 HALF-FACE MASK WITH ASSISTED VENTILATION	Sur commande
CLEANSFLO 2 FULL-FACE MASK WITH ASSISTED VENTILATION	Sur commande
A1P3R FILTER FOR CLEANSFLO 2	Sur commande
CLEANSFLO MASK WITH HIGH-PERFORMANCE ASSISTED VENTILATION	Sur commande
P3R FILTER FOR CLEANSFLO ULTRA	Sur commande
VERSAFLO MASK WITH ASSISTED VENTILATION	Sur commande
VERSAFLO WELDER MASK WITH ASSISTED VENTILATION	Sur commande
P3R FILTER FOR VERSAFLO	Sur commande
ABEKP3R FILTER FOR VERSAFLO	Sur commande

PPE	CODE
COMPACT AIR VENTILATED MASK	EP001794
HALF MASK COMPACT AIR	EP008411+taille (M ou L)
P3 FILTER FOR COMPACT AIR	EP005089
TH1 VENTILATED POWERCAP	EP010346
HARNES	
TRACTEL HT55BY HARNES	EP010818+taille (S, M, XL)
HONEYWELL RM HARNES	EP008324+taille (S/XS, M/L)
TRACTEL HT ELECTRA HARNES	À venir
RUNNER 150cm	EP000076
IK 208 2m positioning lanyard	EP002710
Fall-arrest lanyard with shock absorber 1.5m Manyard ME68	EP002715
1.8m automatic fall-arrestor shock absorber	EP000079
10m automatic fall-arrestor	EP000080
RK1 karabiner with automatic locking	EP003260
2m Turbolite automatic fall-arrestor	EP000081 ou EP000083 ou EP000084

Pablo codes

PPE	CODE
SPECIFIC	
WELDER'S APRON 90cm or 110cm	EP005447 ou EP005448
WELDER'S JACKET	EP004591
WELDER'S GAITERS	See previous pages
WELDER'S HOOD	See previous pages
WELDER'S GLOVES	See previous pages
WELDER'S GLASSES	See previous pages
WELDER'S MASK	See previous pages
CUT-RESISTANT LEGGINGS	EP010571
LEGGINGS EXTENDER	EP010572
TYVEK DISPOSABLE COVERALLS WITH THUMB LOOP	Multiple choices
MICROGARD 2000 DISPOSABLE COVERALLS WITH THUMB LOOP	Multiple choices
3M WHITE DISPOSABLE COVERALLS	Multiple choices
3M BLUE/WHITE DISPOSABLE COVERALLS	Multiple choices
TYPE 5 WHITE DISPOSABLE COVERALLS	Multiple choices
TYPE 4-5-6 WHITE DISPOSABLE COVERALLS WITH SELF-ADHESIVE FLAP	Multiple choices
TYVEK TYPE 5-6 WHITE DISPOSABLE COVERALLS	Multiple choices
TYVEK TYPE 5-6 BLUE DISPOSABLE COVERALLS	Multiple choices
MANUAL PILOT 165 LIFE VEST	EP007562
AUTOMATIC PILOT 165 LIFE VEST	EP008699
LIFE BUOY Ext. diam. 75cm	EP007193

PPE	CODE
O2 SINGLE-GAS DETECTOR, CO, CO ₂ or H ₂ S	EP005031 (CO), EP005032 (H ₂ S) EP005033 (O ₂)
ALTAIR MULTI-GAS DETECTOR: 4X: O ₂ + H ₂ S + CO + EXPLOSIVE GASES	EP005034
ABEK MINISCAPE EMERGENCY ESCAPE HALF-FACE MASK (5min)	EP003245
ABEK MINISCAPE EMERGENCY ESCAPE HALF-FACE MASK (15min)	EP005029
ROLL OF AMERICAN OR ORANGE ADHESIVE FABRIC TAPE	CO004017
ANTI-SLIP OVERBOOTS	Code à venir
CLEANING WIPES	EN007450
POWERFLOW FULL-FACE MASK	See previous pages
CLEANSFACE MASK WITH HIGH-PERFORMANCE ASSISTED VENTILATION	See previous pages
NATURAL LATEX GLOVES	See previous pages
SUMMER COOLING COWL FOR CROWN OF HEAD AND NECK	EP008459
MESH WAISTCOAT	EP010559OR+taille (S/M à 2/3XL)
SUMMER WAISTCOAT	EP009636
SUN CAPE NAPE PROTECTOR ORANGE UV ABSORBANT	EP008003
FLEECE UNDER-HELMET BONNET + HV NECK PROTECTOR + MUFFLER	EP008002 + EP008001 + EP008000
WATER-REPELLENT UNDER-HELMET HOOD	EP005664
COLD-RESISTANT GLOVE LINERS	EP005110
WINTERSAFE COLD-RESISTANT GLOVES IC5	EP010576+taille (8 à 10)
NON-SLIP SOLES	EP005425+pointure (37/41 à 44/48)
LINER FOR COSMOS BOOTS	EP008410GS+pointure - 26/37 à 48/49)
FLEECE NECK WARMER	EP010072
SOCKS	EP017026+taille (39/43 ou 44/48)
PROTECTIVE COLD-WEATHER BONNET	EP005552GS+taille (S à L)



NORME EN 397

MANDATORY REQUIREMENTS

- Shock absorbent (5 kN)
- Penetration resistant
- Flame resistant
- Anchor point for chin strap (breaking force between 150 N and 250N)

OTHERS REQUIREMENTS

- Strength and penetration tests at very low and very high temperatures (-20°C, -30°C, +150°)
- Electrical insulation test
- Lateral deformation
- Splashes of molten metal

NORME EN 812

MANDATORY REQUIREMENTS

- Shock absorbent
- Penetration resistant
- Anchor point for chin strap (breaking force between 150 N and 250 N)

OTHERS REQUIREMENTS

- Strength and penetration tests at very low and very high temperatures (-20°C, -30°C, +150°)
- Electrical insulation test
- Flame resistant

The wording "Warning: this is not a protective helmet" must appear on the shell.

THE HELMET'S PERIOD OF USE AND PROPERTIES OF ITS MATERIALS

The lifetime of a protective helmet depends on its constituent materials, its use and its maintenance (indicated in the manufacturer's manual),

It should be inspected regularly. It must be scrapped if its condition changes (e.g. colour, damage, deformation, cracks or defibering) or after an impact. The expiry date or usage period is given in the user guide that must be supplied with each product.

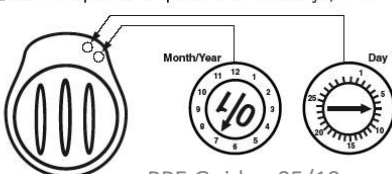
The date shown in the mandatory marking on each helmet is the date it was manufactured. Only helmets marked NF show a usage end date as shown below:

MATERIAL	POLYETHYLENE	POLYAMIDE	ABS	PHENOL TEXTILE	GLASS-FIBRE REINFORCED POLYESTER
Period of use	36 months	48 months	48 months	60 months	60 months
Resistance to ageing	Good	Good	Good	Excellent	Excellent
Resistance to UV	Acceptable	Average	Acceptable	Excellent	Excellent
Resistance to heat deformation	Up to 70°C	Up to 150°C	Up to 90°C	Up to 500°C	Up to 500°C
Melting point	150°C	220°C	180°C	Carbonization at 1000°C	Carbonization at 1000°C
Resistance to very cold conditions	Very good Limit -40°C	Average Limit -20°C	Good Limit -30°C	Excellent with no limit	Excellent with no limit
Resistance to chemicals	Good except oil and grease	Acceptable except for acids and bases	Acceptable except for acids	Good	Good

Location #1: This example shows a cap that was molded on January 3, 2006.



Location #2: This example shows a cap that was molded on July 7, 2001.



For more information, see PPE Guide Specifications



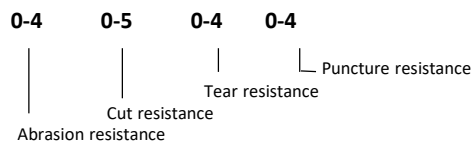


STANDARDS / RECOMMENDATIONS



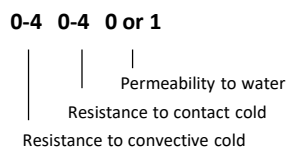
**Mechanical
Risks
EN 388**

Performance level



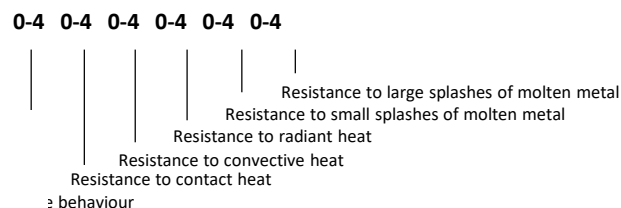
**Cold
EN 511**

Performance level



**Thermal
hazard
EN 407**

Performance level



Performance level	1	2	3	4
A Resistance to flammability (time in seconds after flame and after glow time)	<20s no requirement	<10s <120s	<3s <25s	<2s <5s
B Contact heat (contact temperature for a minimum period of 15s)	100°C	250°C	350°C	500°C
C Convective heat (heat transfer delay)	>4s	>7s	>10s	>18s
D Radiant heat (heat transfer delay)	>7s	>20s	>50s	>95s
E Small drops of molten metal (number of drops)	>10	>15	>25	>35
F Large quantities of molten metal (mass)	30g	60g	120g	200g

Performance level	1	2	3	4
A Convective cold, thermal insulation: I_{TR} in m^2K/W	$0.10 < I_{TR} < 0.15$	$0.15 < I_{TR} < 0.22$	$0.22 < I_{TR} < 0.30$	
B Contact cold, thermal resistance R in m^2K/W	$0.025 \leq R < 0.050$	$0.050 \leq R < 0.100$	$0.100 \leq R < 0.150$	$0.150 \leq R$
C Permeability to water (for at least 30 min)	Successful			

Level X: the test does not apply or the glove was not tested

For more information,
see PPE Guide Specifications



NEW

The standard on anti-cut protective gloves has changed (ISO 13997 added):

- Better guaranteed anti-cut protection with new tests carried out during manufacture (blade pressure tested in addition to cyclical passes over the glove)
- A new marking on gloves

Abrasion		Tear		Performance letter	NEW
E.g.	4	3	4	3	C (P)
Cutting		Puncturing		Impact	
The cutting index is not necessarily indicated if the performance letter is given (in that case, an X would replace the number)		From A to H An X indicates that the test has not been carried out (i.e. gloves that don't use the blade during the test)		Not indicated if not tested	

Which letter should I use?

**For BYCN:
Index 5 = D or +**

From 21/04/2019, performance letters must be identified on gloves. Today, some suppliers are already applying the standard (on the glove or on the label on the glove).

What should I do with old gloves?

Don't panic! Gloves that were on the market before 21 April 2019 may still be sold until 2023, and will still be valid, so you can continue to use them.

For more information,
see PPE Guide Specifications



STANDARDS / RECOMMENDATIONS

STANDARD EN 374: 2003 – GLOVES GIVING PROTECTION FROM CHEMICALS AND MICRO-ORGANISMS (this standard specifies the capability of gloves to protect the user against chemicals and/or micro-organisms)

PICTOGRAM

DEFINITION AND CRITERIA

PENETRATION

Penetration occurs when a chemical and/or a micro-organism moves through porous materials, seams, micro-perforations or other defects in the material of a protective glove, at a non-molecular level.

Performance level	Acceptable quality level	Inspection level
Level 3	< 0,65	G1
Level 2	< 1,5	G1
Level 1	<4,0	S4

The ‘Chemical resistant’ glove pictogram must be accompanied by a 3-digit code. The code refers to the letters identifying three chemicals (from a pre-defined list of 12 standard chemicals) for which the breakthrough time is at least 30 minutes. The impermeable section of the glove must be at least as long to the minimum length of the gloves specified in Standard EN 420. Penetration: a glove must not leak when it is subject to a test of impermeability to air and/or water. The glove will be tested and inspected in accordance with acceptable quality standards.

Code letter	Chemical
A	Methanol
B	Acetone
C	Acetonitrile
D	Dichloromethane
E	Carbon disulphide
F	Toluene
G	Diethylamine
H	Tetrahydrofuran
I	Ethyl acetate
J	n-Heptane
K	Caustic soda 40%
L	Sulphuric acid 96%

PERMEABILITY

Gloves’ plastic and rubber films are not always impermeable to liquids. In some cases, they may be spongy: thus they absorb liquids and keep them in contact with the skin. It is therefore necessary to measure the transit time, or the time required for a hazardous liquid to come into contact with the skin. Each chemical tested is classed depending on the transit time (performance levels from 0 to 6)

Measured breakthrough time	Protection index
>10 minutes	Class 1
>30 minutes	Class 2
>60 minutes	Class 3

Measured breakthrough time	Protection index
>120 minutes	Class 4
>240 minutes	Class 5
>480 minutes	Class 6

The “General chemical protection” or “Impermeable” pictogram shall be used for impermeable gloves that do not offer chemical resistance or if the level required for gloves protecting specifically against chemicals is not reached.

The “Micro-organism” pictogram shall be used when the glove has reached at least performance level 2 in the penetration test.

Warning: the chemical data do not necessarily reflect the actual duration in the workplace.



GUIDE TO CHEMICAL RESISTANCE

This table provides only general information about the materials. You should bear in mind that a glove's resistance is influenced by factors such as the exact nature of the chemical, its temperature, its concentration, the thickness of the glove and the immersion time, etc.

We advise you to refer to the information on the chemical resistance of each glove and to carry out a preliminary test to determine if the glove is suitable for use in the working conditions.

Key: P=Poor, F=Fair, G=Good, E=Excellent, NR=Not Recommended

Chemical	Neoprene	Nitrile	Latex	PVC	Chemical	Neoprene	Nitrile	Latex	PVC
Acetaldehyde	E	P	F	NR	Kerosene	E	E	P	F
Acetic Acid	E	G	G	F	Lactic Acid	E	E	E	E
Acetone	G	NR	G	NR	Lauric Acid	E	E	G	F
Acetonitrile	F	NR	F	NR	Linoleic Acid	E	E	P	G
Ammonium Hydroxide<30%	E	E	G	E	Linseed Oil	E	E	P	E
Amyle Acetate	NR	E	F	P	Maleic Acid	E	E	P	G
Amyl Alcohol	P	G	G	NR	Methyl Acetate	G	P	P	NR
Aniline	G	NR	P	F	Methyl Alcohol	E	E	E	G
Animal Fats	E	E	P	G	Methylamine	G	E	E	E
Battery Acids	E	E	G	E	Methyl Bromide	NR	NR	NR	NR
Benzaldehyde	NR	NR	F	NR	Methylene Chloride	NR	NR	NR	NR
Benzene	NR	P	NR	NR	Methyl Cellulosolve	E	F	P	-
Benzoyl Chloride	NR	NR	P	NR	Methyl Ethyl Ketone (MEK)	G	NR	G	NR
Butane	F	E	P	P	Methylisobutyl Ketone	NR	P	F	NR
Butyl Acetate	NR	F	P	NR	Methyl Methacrylate	NR	P	P	NR
Butyl Alcohol	E	P	E	G	Mineral Oil	E	E	P	F
Butyl Cellulosolve*	E	E	E	NR	Mineral Spirits	G	E	NR	F
Carbon Acid	E	P	P	G	Monoethanolamine	E	E	G	E
Carbon Disulfide	NR	NR	NR	NR	Morpholine	P	NR	G	NR
Carbon Tetrachloride	P	G	NR	NR	Muriatic Acids	E	G	G	G
Castor Oil	E	E	E	E	Naptha V.M & P.	G	E	NR	P
Cellosole Acetate	E	G	G	NR	Nitric Acid <30%	E	P	G	G
Cellosole Solvent	E	G	E	NR	Nitrile Acid 70%	G	NR	F	F
Chlorobenzene	NR	NR	NR	NR	Nitrile Acid Red Fuming	NR	NR	P	P
Chloroform	F	F	NR	NR	Nitrile Acid White Fuming	NR	NR	P	P
Chloronaphalens	NR	F	NR	NR	Nitrobenzene	NR	NR	P	NR
Chloroethene VG	NR	F	NR	P	Nitromethane	E	F	G	P
Chromic Acid	F	F	NR	G	Nitropropane	G	NR	E	NR
Citric Acid	E	E	E	E	Octyl Alcohol	E	E	G	F
Cottonseed Oil	E	E	P	G	Oleic Acid	E	E	P	F
Cresols	G	G	P	F	Paint Remover	G	G	F	P
Cutting Oil	E	E	F	P	Palmitic Acid	E	G	G	G
Cyclohexane	F	E	P	P	Pentachlorophenol	E	E	P	F
Cyclohexanol	E	E	P	G	Pentane	E	E	P	NR
Dibutyl Phthalate	F	G	P	G	Perchloric Acid 60%	E	E	P	E
Diethylamine	P	F	NR	NR	Potassium Hydroxide <50%*	E	G	E	E
Di-Isobutyl Ketone	P	E	P	P	Printing Ink	G	E	G	F
Dimethyl Formamide (DMF)	G	NR	E	NR	Propyl Acetate	P	F	P	NR
Dimethyl Sulfoxide (DMSO)	E	E	E	NR	Propyl Alcohol	E	E	E	F
Dicotyl Phthalate (DOP)	G	G	P	NR	Perchloroethylene	NR	G	NR	NR
Dioxane	NR	NR	NR	NR	Phenol	E	NR	G	G
Ethyl Acetate	F	NR	P	NR	Phosphoric Acid*	E	E	G	G
Ethyl Alcohol	E	E	E	G	Picric Acid	E	E	G	E
Ethylene Dichloride	NR	NR	P	NR	Propylene Oxide	NR	NR	P	NR
Ethylene Glycol	E	E	E	E	Rubber Solvent	G	E	NR	NR
Ethyl Ether	E	E	NR	NR	Sodium Hydroxide <50%	E	G	E	G
Ethylene Trichloride	P	P	P	NR	Stoddard Solvent	E	E	P	NR
Formaldehyde	E	E	E	E	Styrene*	NR	NR	NR	NR
Formic Acid	E	F	E	E	Sulfuric Acid 95%	F	G	NR	NR
Freon	G	F	NR	NR	Tannic Acid	E	E	E	E
Furfural	G	NR	E	NR	Tetrahydrofuran (THF)	NR	NR	NR	NR
Gasoline	P	E	NR	P	Toluene	P	G	NR	NR
Glycerine	E	E	E	E	Toluene Di-Isocyanate (TDI)	NR	NR	P	P
Hexane	E	E	NR	NR	Trichlorethylene (TCE)	P	G	NR	NR
Hydraulic Fluid Petro. Based	F	E	P	G	Tricresyl Phosphate (TCP)	F	E	G	F
Hydraulic Fluid Easter Based	P	P	P	P	Triethanolamine 85% (TEA)	E	E	G	E
Hydrazine 65%	E	E	G	E	Tung Oil	E	E	NR	F
Hydrochloric Acid*	G	E	E	E	Turbine Oil	E	G	P	F
Hydrofluoric Acid	G	E	E	E	Turpentine	G	E	P	P
Hydrogen Peroxide	E	E	E	E	Vegetable Oil	E	E	P	F
Hydroquinone	G	E	E	E	Xylene	P	G	NR	NR



STANDARDS / RECOMMENDATIONS

WHICH MATERIALS? (for reusable gloves)

NATURAL LATEX

Natural rubber (hevea): the most elastic of all substances known

Good resistance to wear and tearing, and to all water-soluble and diluted products. Also available in disposable gloves.

Poor resistance to oily or greasy products and to hydrocarbons. May cause allergic reactions.

NEOPRENE

Synthetic rubber based on polychloroprene

Good resistance to strong acids and bases.

Average mechanical strength Non resistant to aromatic or chlorinated solvents.

PVA

(Polyvinyl alcohol) Synthetic polymer based on vinyl alcohol

Excellent resistance to many hydrocarbons (aliphatic, aromatic and chlorinated, etc.), to esters and to ketones

Breaks down on contact with water. Expensive

FLUORINATED MATERIALS

Synthetic materials e.g Viton® or Téflon® brands

Good resistance to many products, including benzene and PCBs

Low resistance to cuts and abrasion. Expensive

NITRILE

Synthetic rubber (acrylonitrile-butadiene-styrene copolymer or ABS)

Good mechanical strength: wide chemical resistance (oils, greases, alcohols and petroleum products, etc.). Also available in disposable gloves.

Poor resistance to ketones and halogenated products (chlorinated or fluorinated, etc.)

PVC

(Polyvinyl chloride) Synthetic polymer based on vinyl chloride, also known as "Vinyl"

Fair resistance to acids, bases and alcohols. Moderate cost. Also available in disposable gloves.

Poor resistance to ketones, aldehydes and to aromatic or halogenated hydrocarbons

BUTYL

Synthetic rubber (isobutylene-isoprene copolymer)

Good resistance to strong acids, ketones, esters, glycol ethers, amines and aldehydes, etc.

Poor resistance to aliphatic aromatic and halogenated hydrocarbons. Expensive

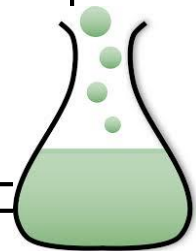
MULTILAYER MATERIALS

Laminated layers of polyethylene and ethylene vinyl alcohol copolymers, brand

names Barrier®, Silver Shield®

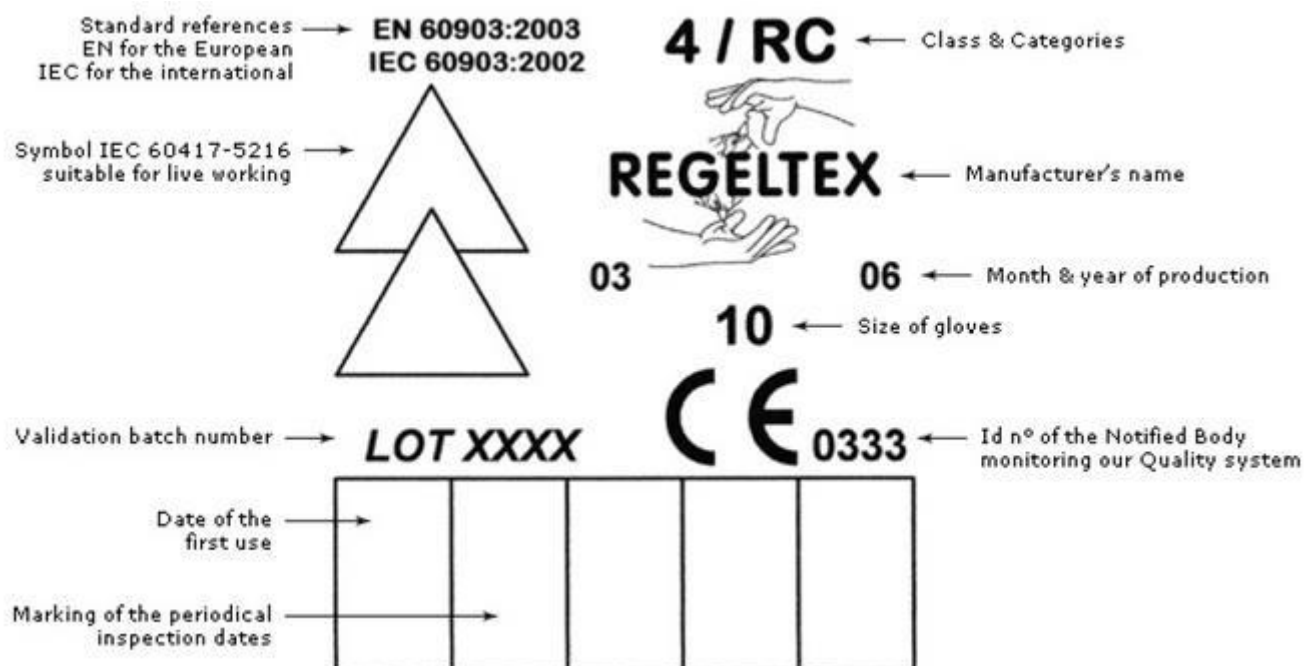
Excellent resistance to most chemicals

Reduces dexterity, poor mechanical strength



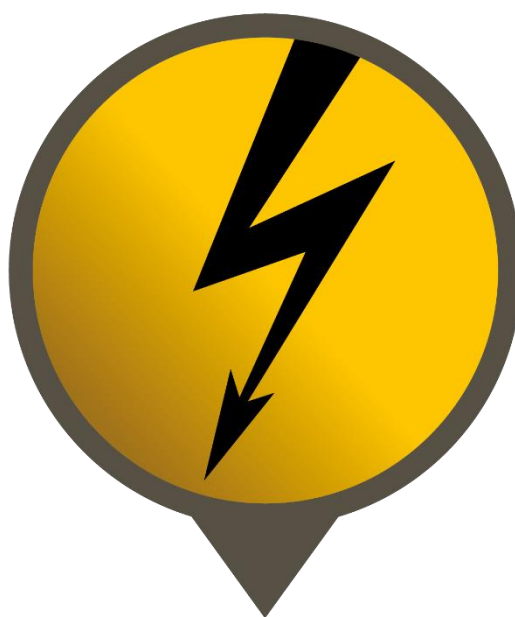


STANDARDS / RECOMMENDATIONS



Class	Maximum thickness (mm)	Test voltage (volts)	Usage voltage (volts)
00	0.5	2500	500
0	1	5000	1000
1	1.5	10000	7500
2	2.3	20000	17000
3	2.9	30000	26500

No gloves, even those kept in stock, shall in principle be used unless it has been checked within the last 6 months.



STANDARDS / RECOMMENDATIONS



	Safety footwear EN ISO 20345/A1	Protective footwear EN ISO 20345/A1	Occupational footwear EN ISO 20347/A1
Class I or II	SB Basic requirements	PB Basic requirements	
Class I (leather or other materials except for all-rubber and all-polymer models)	S1 Basic requirements - Closed heel - Antistatic - Energy absorption around heel S1P as for S1 + - Puncture-resistant sole	P1 Basic requirements - Closed heel - Antistatic - Energy absorption around heel	O1 Basic requirements - Closed heel - Hydrocarbon-resistant sole - Antistatic - Energy absorption around heel
	S2 as for S1 + - Impermeability to water		
	S3 as for S2 + - Puncture-resistant sole - Cleated sole		
Class II (all-rubber and all-polymer footwear)	S4 Basic requirements - Antistatic - Energy absorption around heel	P4 Basic requirements + - Antistatic - Energy absorption around heel	O4 Basic requirements + - Antistatic - Energy absorption around heel
	S5 as for S4 + - Puncture-resistant sole - Cleated sole		

A	Antistatic
C	Conducting footwear
CI	Insulation against cold
CR	Cut-resistant upper
E	Energy-absorbing heel
HI	Insulation against heat
HRO	Sole resistant to contact heat
P	Puncture-resistant sole
FO	Hydrocarbon (fuel)-resistant sole
WRU	Upper resistant to water absorption and penetration
AN	Malleoli (ankle) protection
WR	Whole shoe waterproof
M	Metatarsal protection
I	Insulating shoe

Slip resistance requirements:

Safety footwear shall fall into one of the categories below:

SRA – Slip resistance on ceramic tiles covered with a solution of sodium lauryl sulphate.

SRB – Slip resistance on a steel floor covered with glycerine.

SRC – Slip resistance on ceramic tiles and on a steel floor.





STANDARDS / RECOMMENDATIONS

LEXICON OF MATERIALS

Various materials are used to make eyepieces and shields

EYEPIECES			SHIELDS	
Polycarbonate	Acetate	Mineral glass	Polyester	Polyamide
Thermoplastic and thermoformable polymer; 40 times more resistant and three times lighter than glass, provides perfect optical quality and good heat resistance (>140°C), filters UV 100%	Resin / cellulose; lighter than glass, average heat resistance (below 100°C), withstands solvents and chemicals well	Produced by fusing minerals, resists impacts, very good chemical resistance	UV filtration Complete electrical insulation, withstands high temperatures well, self-extinguishing	Good thermal and electrical insulation

Marking on eyepieces

SYMBOLS FOR MECHANICAL STRENGTH	
A	High-energy impact (strength test with a 6mm sphere weighing 0.86g at 190 m/s)
B	Medium-energy impact (strength test with a 6 mm sphere weighing 0.86g at 120 m/s)
F	Low-energy impact (strength test with a 6 mm sphere weighing 0.86g at 45 m/s)
S	Increased robustness (strength test with a 22 mm sphere weighing 43g falling from a height of 1.30m)
K	Resistance to surface damage from fine particles (option)
N	Resistance to fogging (option)
T	Resistance to mechanical impacts at extremes of temperature (-5°C/+55°C)

OCCUPATIONAL RISKS

The eyes shall be protected against certain occupational risks

RISKS		CONSEQUENCES FOR THE EYES
Mechanical	Impacts, dust, solid particles, grains of sand	Corneal lesions or perforations, tears in the iris, clouding of the lens
Chemical	Droplets and splashes of liquid: acids, bases, solvents and contaminating liquids	Corneal burns or disorders, acute conjunctivitis, ulcers, viral infections
Thermal	Hot liquids, molten materials, flames	Destruction of the eye
Electrical	Direct contact, electrical arc from a short-circuit	Retinal burns, corneal lesions, damage to the lens
Linked to radiation	Ultraviolet, infrared, laser, very strong light	Inflammation of the conjunctiva, cataract, retinal lesions or burns



STANDARDS / RECOMMENDATIONS

WELDING FILTERS FOR ELECTRIC ARCS

PASSIVE WELDING FILTERS

Mineral glass for welding
Mineral glass for welding marked as heat resistant
Dim. (in mm) : 105x50 – 105x30 – 110x90
Shades: 8 - 14

COVER PLATES

Clear mineral glass
Clear non-stick C39
Dim. (in mm) : 105x50 – 105x30 – 110x90

		A M P S																							
		0.5	1	2.5	5	10	15	20	30	40	60	80	100	125	150	175	200	225	250	275	300	350	400	450	500
	Stick								9	10	11				12				13						
	Mild Steel Mig with Argon												10	11	12				13						
	Mild steel Mig with CO ₂									10	11	12	13												
	Aluminum Mig												10	11	12		13								
	Tig					9	10	11				12	13												
	FCAW														10	11	12	13							
	Plasma Cutting												11		12		13								
	Plasma Welding	4	5	6	7	8	9	10																	

This table indicates the recommended darkness of shade best suited to filters for different types of welding procedure. Depending on the conditions of use and the welder's preference, the figure on either side may be selected from the scale.



STANDARDS / RECOMMENDATIONS

For the welding filters, the scale number refers only to the protection class. Depending on the conditions of use, you may use a filter with a number one place up or down on the scale, but it may be dangerous to use filters with too high a scale number (too dark) as this encourages the operator to approach the radiation source and breath in noxious fumes.

Scale numbers for gas welding and braze welding

	q = flow of acetylene in litres per hour			
	q<70	70<q<200	200<q<800	q>800
Welding and braze welding	4	5	6	7

Work	q = flow of acetylene in litres per hour		
	900<q<2000	2000<q<4000	4000<q<8000
Oxygen cutting	4	5	6





STANDARDS / RECOMMENDATIONS

STANDARDS	
EN 352-1	Individual hearing protectors. General requirements. Relating to ear muffs.
EN 352-2	Individual hearing protectors. General requirements. Relating to ear plugs and headbands.
EN 352-3	Individual hearing protectors. General requirements. For ear muffs fixed to protective helmets used in industry.
EN 352-4	Individual hearing protectors. General requirements. Relating to level-dependent protectors.
EN 458	Recommendations for selection, use, precautions during use and maintenance of personal protective equipment.

*For more information,
see PPE Guide Specifications*



STANDARDS / RECOMMENDATIONS

STANDARDS	
EN 132	Definitions of terms
EN 133	Classification
EN 135	List of equivalent terms
EN 137	Self-contained open-circuit compressed air breathing apparatus
EN 139	Fresh air hose breathing apparatus for use with full or half face mask
EN 140	Half masks and quarter masks
EN 141	Replaced by EN 14387
EN 143	Particle filters (negative-pressure breathing apparatus)
EN 146 and 147	Assisted-ventilation filtering apparatus to protect against particles
EN 149	Filtering half masks to protect against particles
EN 270 and 271	Compressed air line breathing apparatus
EN 371	Replaced by EN 14 387
EN 404	Respiratory protective devices for self-rescue
EN 405	Valved filtering half masks to protect against gases or gases and particles
EN 12941	Respiratory protective devices. Powered filtering devices incorporating a helmet or a hood
EN 14387	Respiratory protective devices. Gas filter(s) and combined filter(s)
EN 12942	Respiratory protective devices. Power assisted filtering devices incorporating full face masks or half masks
EN 14594	Respiratory protective devices. Continuous flow compressed air line breathing apparatus



For more information,
see PPE Guide Specifications



STANDARDS / RECOMMENDATIONS

	Aerosol type	Main applications	Limit
Class 1 (P1 or FFP1)	Large solid particles, not specifically toxic	Small amounts of fine dust and aqueous or oily mist, generally from sandblasting, drilling and cutting	4 x TWA
Class 2 (P2 or FFP2)	Solid and/or liquid aerosols indicated as irritants or hazards	Moderate amounts of fine dust and aqueous or oily mist, generally from plastering, cementing or sandblasting; and dust from soft wood	10 x TWA
Class 3 (P3 or FFP3)	Toxic solid and/or liquid aerosols	Large amounts of fine dust and aqueous or oily mist, generally from handling toxic powders present in the pharmaceutical industry or from jobs involving biological agents and fibres	With half-face mask: 50 x TWA With full-face mask: 200 x TWA

■ FILTERS AGAINST GASES AND VAPOURS

There are 3 classes of filters:

Class 1: low capacity (pad)

Class 2: medium capacity (cartridge)

Class 3: the highest capacity (drum)

The filter is marked for the type (A, B etc.) and class (1, 2 or 3)

TYPE OF FILTER	COLOUR	FIELD OF USE
A	Brown	Organic gases and vapours – boiling point > 65°C
AX	Brown	Organic gases and vapours – boiling point < 65°C
B	Grey	Inorganic gases and vapours except CO
E	Yellow	Acid gases and vapours
K	Green	Ammonia and its derivatives
P	White	Particles, solid and liquid aerosols
Hg	Red	Mercury vapour
NO	Blue	Nitrous gases and nitrogen dioxide

■ RESTRICTIONS ON USE

Pad or cartridge filters shall not be used when the oxygen concentration is below 17% by volume.

After each use, the apparatus shall be cleaned, disinfected and prepared for subsequent reuse.



Written by G Robin
COPIL PPE BYCN 2019

